When the query is executing, the state is "active". When the query is no longer executing, a state_change timestamp is recorded. For example, when a connection is no longer active and is now idle, state_change should be set at the point where that transition occurred.

An idle connection is available for re-use by the client once it is no longer performing a transaction.

If connections are idle, how can those be managed?

Managing Idle Connections

The idle_timeout timeout in the Active Record Connection Pool can be set to determine a maximum allowable time that idle connections can be checked out from the pool. When they are idle and checked out for longer than the max allowed value, those connections will be forcibly returned to the pool. By default the value is 300 seconds (5 minutes).

A similarly named PostgreSQL server parameter exists called idle_session_timeout[3] which is new in Version 14. The parameter has this description:

> Terminate any session that has been idle (that is, waiting for a client query), but not within an open transaction, for longer than the specified amount of time.

This parameter can complement the idle_timeout parameter in the Active Record Connection Pool. The server parameter provides an additional layer of protection within PostgreSQL from connections that are idle and not completing any work. The reason those are bad is because they lock up one of the limited database connections.

Idle connections that are open for a long time may be terminated. For example, for pid 87581 with a idle connection and a query that's not actively running, cancel it using a query like this one:

sql/pg_cancel_backend.sql

```
-- cancel pid 87581
SELECT pg_cancel_backend(87581);
```

In *Waiting for PostgreSQL 14 – Add idle_session_timeout*, experiments are conducted with idle_session_timeout to help demonstrate how it works.

A query is sent to the background and then the postgresql.log shows the terminated connection with error FATAL: terminating connection due to idle-session timeout after the maximum time was exceeded. In the second example, despite

3. https://postgresqlco.nf/doc/en/param/idle_session_timeout/